Gazint IntelliScript

Standard Operating Procedure

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Purpose

Gazint IntelliScript is a hardware IT automation tool that emulates a USB keyboard and mouse to execute administrative input on a workstation. This document outlines the authorization, configuration, operating, and safeguarding requirements for IntelliScript devices.

What is Gazint IntelliScript?

IntelliScript is a programmable input device built on an ATmega32U4. The ATmega32U4 provides native USB support, which allows the board to operate as Human Interface Device. Pressing a button on the device executes a preprogrammed sequence of administrative actions onto the workstation the device is connected to.

Device Overview

The following describes IntelliScript's configuration.

- Controller: ATmega32U4
- HID libraries: HID-Project BootKeyboard and BootMouse
- Inputs: Button A on pin 16 & Button B on pin 15.

Idle Screen

When the device has power and an administrate action is not being executed, the OLED displays the idle image shown below.

IntelliScript Usage Requirements

Authorization

- Connect IntelliScript only to workstations you are already authorized to access and administer.
- Do not connect the device to a workstation belonging to another user, a customer, or a third party without documented approval from Leadership.

Personnel and Training

- Only personnel trained on this SOP and on the device's current payloads may operate IntelliScript.

- Know exactly what each button does before connecting the device. Do not press a hotkey to find out what it does.
- Personnel who have not reviewed the documented hotkey behavior within the designated SharePoint file, are not qualified to operate the device.

Storage and Custody

- Store it in a secure and authorized storage location when not in use.
- Asset tag the device so it is identifiable as IT equipment and not an unknown USB device.

Using IntelliScript on a Workstation

Prepare the Workstation

- Confirm you are at the correct workstation and that the work you are about to automate is authorized.
- Save or close unrelated work. IntelliScript input is delivered to whichever window currently has focus.
- If the task requires the BIOS/UEFI screen, bring the workstation to that screen before continuing.

Connect the Device

- Connect IntelliScript to an available USB port on the target workstation.
- Wait for the OLED to display the idle image, which indicates the device has power and the sketch is running.

Trigger the Hotkey

- Verify that the intended target is correct.
- Press and release the button for the sequence you intend to run.
- Do not press a second button while a sequence is running.
- Watch the workstation while the sequence executes. If it produces unexpected output, disconnect the device immediately.

Disconnect and Return

- Disconnect the device once the task is complete. Do not leave it connected to an unattended workstation.
- Verify the workstation is in the expected state before leaving it.
- Return the device to the authorized storage location and record the return in a custody log.

Risks of Usage

Native Keyboard Override and Masked Keyboard Failures

IntelliScript supplies keyboard and mouse inputs that the workstation treats exactly as it would input from the built-in or attached keyboard. During hardware troubleshooting this can mask a genuine keyboard failure.

Stored Credentials in BIOS Hotkeys

A hotkey that unlocks a BIOS/UEFI screen contains the BIOS password as literal text. That value exists in plaintext in the sketch source and in the compiled firmware on the board. Anyone who obtains the device, the source file, or a firmware dump obtains the password. A device programmed this way is a credential, and the exposure extends to every workstation that shares that BIOS password.

Loss, Theft, or Unattended Devices

The device is small enough to be misplaced and portable enough to be taken. Authorized personnel, secured storage locations and custody logs mitigate this risk.

Unreviewed Code and Firmware Changes

IntelliScript's behavior is defined entirely by the code loaded onto the board. Anyone able to connect the board to a computer running the Arduino toolchain can re-flash it, and the OLED image gives no indication that the payload behind a button has changed. An unreviewed edit, an untested change, or a re-flash by an unauthorized party alters what the device does with no visible sign at the point of use.

Safe Practices

Verifying Keyboard Function

- When diagnosing a suspected keyboard failure, disconnect IntelliScript before testing.
- Test the workstation's own keyboard manually and confirm it produces input on its own.
- Do not close or downgrade a keyboard-related ticket on the basis of input delivered through IntelliScript.

Handling Devices Programmed with a BIOS Password

- Treat a device as a sensitive asset.
- Store it in a secure storage location. Do not leave it in a desk drawer, a bag, or a shared toolkit.
- Do not lend, hand off, or leave the device unattended while it holds a BIOS payload.
- If the device is lost or its custody becomes uncertain, treat the BIOS password as compromised and initiate a password change.

Physical Custody

- Keep the device either on your person or in authorized storage locations. There is no acceptable state in between.
- Disconnect and store the device before stepping away from a workstation, even briefly.
- Log check-out and returns into the custody log.
- Device must be asset tagged and added to the ServiceNow alm.assets table.
- Report a lost or stolen device to Leadership immediately, and state what payload the device was flashed for.
- Reconcile all IntelliScript devices against the custody log.

Code and Firmware Change Control

- Only authorized personnel may edit the code or re-flash a device.
- Maintain the sketch in secured sharepoint folder location, with a version identifier.
- Have a two qualified personnel review every payload change before it is flashed.
- Test every change on a non-production workstation before using the device on production or user-facing equipment